



Formation AgI and ZnI₂ nanocrystals in AgI-ZnI₂-SiO₂ hybrid powders

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Abstract.

AgI and ZnI₂ nanocrystals were key components for AgI-ZnI₂-SiO₂ hybrid powders (HPs), which could be potentially important for atmospheric artificial precipitation technology. HPs were created by the "Hydrothermal template cocondensation" method ("HTC" method). Mesoporous silica dioxide (MCM48, MCM41, SBA15, SBA16), silver iodides and zinc iodides were simultaneously grown under specific conditions. Influence of silica dioxide on AgI and ZnI₂ nanocrystals characteristics (phase, size and thermal stability) were studied by various physicochemical analysis methods. In addition to crystal features, some structural, textural properties AgI-ZnI₂-SiO₂ hybrid as individual agglomerate and its morphology were determined. It was showed that nanocrystal features were depended on synthesis condition. The "chemical pressure" exerted by the pH-former agent had an impact on the initial stages of the process, while the morphology of the silica dioxide matrix controlled the crystal properties during the post-synthesis phase. The silica dioxide morphology controlled of crystal properties on the post-synthesis stage. It was established that thermal stability of AgI and ZnI₂ nanocrystals increased due to protective shielding function of that SiO₂ matrix.

Keywords: hybrids; silver iodide; artificial technology; silica dioxide

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1. Introduction

The technology of atmospheric artificial precipitation is strategic sphere to prevent hail, drought and heavy rains. Silver iodide is the most well-known substance for "cloud seeding". But it has a number of disadvantages, namely high crystal size, polymorphism of crystal lattice, low thermal stability. These parameters are crucial, because silver iodide does not contact with ice crystal. Firstly, then silver iodide heated into delivery vehicles, it destroyed to metal silver and silver oxide. Secondly, large native AgI crystal cannot interact with icicle. There are common approaches to correct these shortcomings.

The first approach is to reduce the silver iodide dispersion by changing the parameters of the chemical synthesis of its crystals. The second way is to prevent AgI destruction by the addition of iodine donors, such as copper iodide, zinc iodide, magnesium iodide or iodate of these metals [1-4].

Previously, the authors created a AgI-SiO₂ hybrid powder hybrid. The main idea of their experimental works was to localize AgI crystals on SiO₂ matrix. The authors took chemical reaction between silver nitrate and potassium iodide as the basis for the formation of silver iodide crystals. The synthesis methods proposed by the authors have

significantly increased the thermal stability of silver iodide and reduced the dispersion of its crystals [5-8].

Since the use of iodine donors also makes it possible to increase the thermal stability of silver iodide, it is necessary to solve problems, which are related to determining the methods of their introduction into the SiO₂ matrix and to establish the influence of synthesis conditions on the characteristic features of AgI and ZnI₂ crystals.

2. Methods or Materials

"Hydrothermal template cocondensation" method ("HTC" method) was the basic way for AgI-ZnI₂-SiO₂ HPs synthesis. The "HTC" method conditions and parameters were established by the authors in previous studies [5-8]. This method allowed to simultaneously synthesize silicon dioxide, silver iodide (from silver nitrate) and zinc iodide (from zinc nitrate) in the presence of excess potassium iodide. Different types of mesoporous silicon dioxide (such as MCM48, MCM41, SBA15 and SBA16) served as a base for the silver iodide and copper iodide crystals.

Some substances were used for the synthesis: tetraethoxysilane (TEOS) (98%, Sigma Aldrich), silver nitrate (99%, Sigma Aldrich), zinc nitrate (99%, Sigma Aldrich), potassium iodide (97%, Sigma Aldrich), cetyltrimethylammonium bromide (CTAB) (98%, Sigma Aldrich), Pluronic P-123, (98%, Sigma-Aldrich), Pluronic F-127, (98%, Sigma-Aldrich), sodium hydroxide (98%, Sigma Aldrich), Ammonium hydroxide solution (28-30 %, Sigma Aldrich), hydrochloric acid (36%, Sigma Aldrich).

AgI-ZnI₂-MCM48 HPs synthesis conditions were parameters: hydrothermal aging time was 48 hours, hydrothermal aging temperature was 120 °C, pH of the sol was 10 - 12. Ratio [TEOS] / [CTAB] was 1 : 0.44, ratio [Ag] / [Si] was 0.08.

AgI-ZnI₂-MCM41 HPs synthesis conditions hybrid were parameters: hydrothermal aging time was 48 hours, hydrothermal aging temperature was 100 °C, pH of the sol was 10 - 12. Ratio [TEOS] / [CTAB] was 1 : 0.20, ratio [Ag] / [Si] was 0.08.

AgI-ZnI₂-SBA15 HPs synthesis conditions were parameters: hydrothermal aging time was 24 hours, hydrothermal aging temperature was 80 °C, pH of the sol was 2-3. Ratio [TEOS] / [Pluronic P-123] was 1 : 0.02, ratio [Ag] / [Si] was 0.08.

AgI-ZnI₂-SBA16 HPs synthesis conditions were parameters: hydrothermal aging time was 24 hours, hydrothermal aging temperature was 80 °C, pH of the sol was 2-3. Ratio [TEOS] / [Pluronic F-127] was 1 : 0.02, ratio [Ag] / [Si] was 0.08 [5-6].

The ratio [Ag] / [Zn] was 1.00 for new AgI-ZnI₂-SiO₂ HPs. Then HPs were washed and dried in a drying oven at a temperature of 110 °C. The next step was the annealing stage. This was a post-synthetic treatment to remove the structure-forming agent (CTAB, Pluronic P-123, Pluronic F-127). It took place in a muffle furnace in an air atmosphere at 550 °C temperature for 5 hours.

The synthesized samples were characterized by various physicochemical methods of analysis.

The structure of the metal iodide crystals and powder materials in general was identified by XRD. Diffraction patterns were recorded by an XRD-7000 diffractometer (Shimadzu, Japan) using CuK α radiation ($\lambda_{av} = 1.54 \text{ \AA}$). The scan ranges were 1.4 -10.0 and 10.0 – 80.0 degrees with a step of 0.010- 0.005 degrees, accumulation time ranged from 1.5 to 2.0 s. The diffraction patterns were identified using the JSPDS powder diffraction file. The average crystallite size was calculated by the Selyakov-Scherrer formula: $D = K \lambda / (\beta \cos \Theta)$, where $K = 0.89$; $\lambda = 1.54056 \text{ \AA}$; β - half-width of the reflection (100), rad; Θ - diffraction angle of the reflection, deg.

Morphological features of the inorganic hybrid samples were examined using a FEI Quanta FEG650 scanning electron microscope (ThermoFisher Scientific, the Netherlands).

The structural and textural properties of HPs were evaluated by low-temperature adsorption of nitrogen with an ASAP 2020 analyzer (Micromeritics, USA) after vacuum degassing of the sample at 350°C for 3 h. The specific surface area of the samples (S_{BET}) and the total pore volume (V_{tot}) were determined by the BET method, pore size distribution was identified from the desorption isotherms by the BJH method in the pore size range of 1.7 – 300.0 nm.

The content of silver iodide and zinc iodide was determined using atomic absorption spectroscopy (Thermo Fisher Scientific iCE 3500, USA) in combination with titration. This approach allowed to clarify the amount of metal iodides in the HPs. Before the measurement, silver and zinc were converted from the HPs into a soluble form. The obtained results were recalculated to values reflecting the direct content of metals in the HPs.

The thermal behavior of HPs was studied using a TGA/DSC 1 device (METTLER-TOLEDO, Switzerland) in an air atmosphere at a heating rate of 10 °C/min in the temperature range of 25–1000 °C.

The study of moisture absorption under static conditions was carried out by saturating the powder material with moisture in a desiccator, where the relative humidity of the environment was maintained at about 90%.

The HPs' activity was investigated under dynamic conditions. The climatic chamber specially was utilized to create an oversaturated wet area and facilitate its precipitation. This climatic chamber was developed by the Institute of Technical Chemistry of the Ural Branch of Russian Academy of Sciences. The chamber was a hermetically sealed, thermally insulated cubic structure with a volume of 8 m³. It was equipped with an ultrasonic humidification system capable of producing 310 cm³/h, a device for spraying HPs, and laser sensors (CDR-10X optical position sensors). The temperature and humidity conditions were regulated and maintained by an air conditioning system within the chamber. The experiments were conducted at a temperature of -5 °C. The duration of the destruction process was calculated by subtracting the time when the hybrid was introduced into the oversaturated wet area from the time it became transparent.

3. Result and discussion

The authors developed some methods for hybrid synthesis and those methods were used for AgI-ZnI₂-SiO₂ powder creation [5-8]. Those methods allowed to AgI-ZnI₂-SiO₂ formation from different silica dioxide matrix: MCM48, MCM41, SBA15, SBA16. The synthesized HPs showed different characteristic features (structural, textural and exploitation properties and thermal stability of nanocrystals) due to different silica dioxide types.

It was established that a microagglomerate was structural unit of powders and samples were polydisperse powder. It was shown that the morphology of the HPs's agglomerates was determined by the characteristics of the native silica dioxide matrix of the corresponding family [9, 10]. SEM results showed that AgI-ZnI₂-MCM48 HP was worm-like agglomerates, AgI-ZnI₂-MCM41 was grape-like agglomerates, AgI-ZnI₂-SBA15, AgI-ZnI₂-SBA16 were elongated spherical agglomerates (fig. 1). SiO₂ type was confirmed by Small-Angle X-ray Scattering. The microagglomerate was a result of stable accumulation of single silica dioxide nanoparticles. It was assumed that electrostatic forces held together silica dioxide nanoparticles. The silver iodide and zinc iodide nanocrystals did not prevent self-assembly of powder microagglomerates.

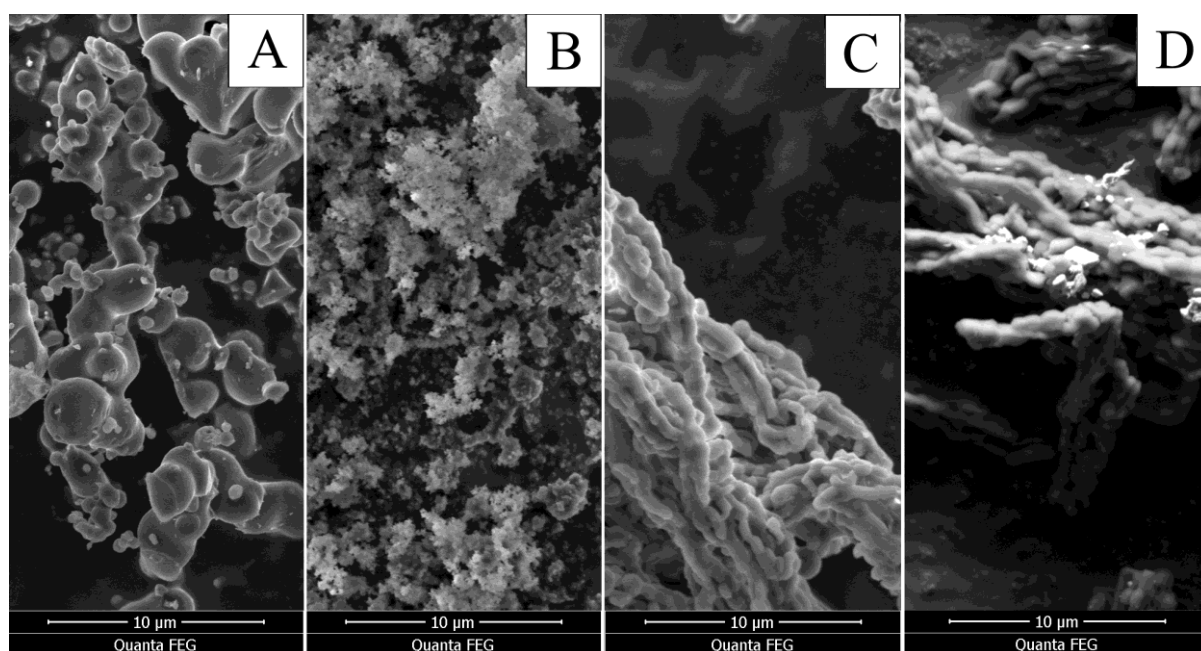


Figure 1. SEM-images of AgI-ZnI₂-SiO₂ HPs: A - AgI- ZnI₂-MCM48, B - AgI- ZnI₂-MCM41, C - AgI- ZnI₂-SBA15, D - AgI- ZnI₂-SBA16

The X-ray phase analysis at low diffraction angles 1-10 ° showed that silica dioxide was mesoporous silica dioxide of MCM and SBA families (fig. 2) [9, 10].

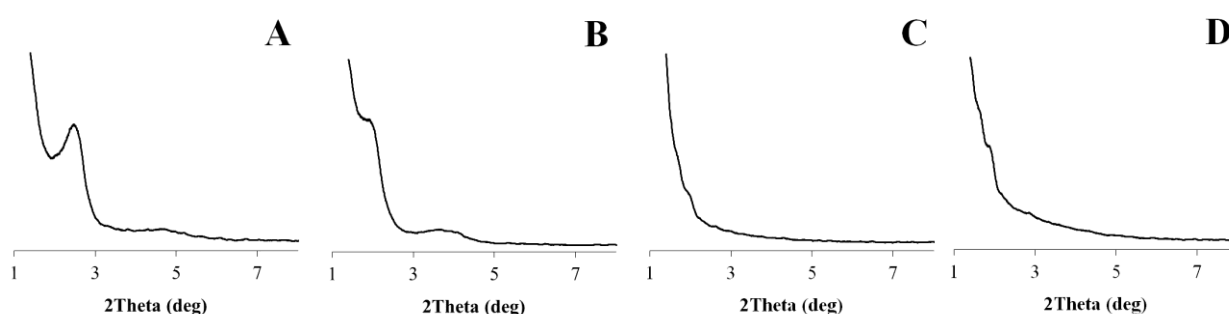


Figure 2. X-ray image of AgI-ZnI₂-SiO₂ HPs in small angles: A - AgI- ZnI₂-MCM48, B - AgI- ZnI₂-MCM41, C - AgI- ZnI₂-SBA15, D - AgI- ZnI₂-SBA16

The SEM and X-Ray results confirmed that silver iodide and zinc iodide did not affect the structure of the silica dioxide matrix.

The low-temperature nitrogen sorption was used for assessment of structural properties of HPs. It was established that a significant decrease in the specific surface area for the synthesized samples compared to the unmodified silica oxide matrix was occurred (Table 1). The decrease was to 20-80 times for this parameter. This might be a consequence of the crystals of silver iodide and zinc iodide were localized on the irregularities (natural bends of the microagglomerate) of the relief, as well as in the pores and interporous space of the silica dioxide matrix. The changes of structural properties occurred for all samples of synthesized AgI- ZnI₂-SiO₂ HPs. It was found that the pore volume indicators and their linear dimensions were significantly changed as well as the specific surface area. It could be because silver iodide and zinc iodide nanocrystals were embedded into the surface and they were the guests with respect to silica dioxide at the stage of the sol existence.

Table 1. The structural and textural properties of AgI-ZnI₂-SiO₂ HPs.

HP sample	Surface area, S_{BET} , m^2/g	Total volume, cm^3/g	Pore diameter, nm (desorption)
AgI- ZnI ₂ -MCM48	40	0,07	6,77
AgI- ZnI ₂ -MCM41	33	0,09	7,12
AgI- ZnI ₂ -SBA15	27	0,11	8,96
AgI- ZnI ₂ -SBA16	25	0,10	9,28
MCM48	1635	0,97	2,6
MCM41	1088	0,84	3,2
SBA15	688	0,72	3,9
SBA16	617	0,67	4,6

Figure 3. was the HPs sorption isotherms. Those isotherms were IV type isotherms (according IUPAC classification). Such types of isotherms described the micro- and mesopores in the HPs samples (fig. 3).

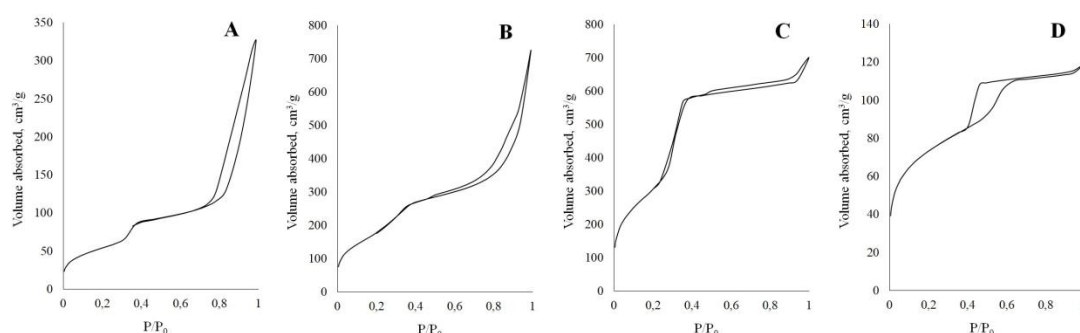


Figure 3. The AgI- ZnI₂-SiO₂ HPs sorption isotherms: A - AgI-ZnI₂-MCM48, B - AgI-ZnI₂-MCM41, C - AgI-ZnI₂-SBA15, D - AgI-ZnI₂-SBA16.

The characteristic features of AgI and ZnI₂ nanocrystals were defined by X-Ray. The Selyakov-Scherrer formula was used for the average crystallite size calculation (Table 2) [11, 12]. It was found that silver iodide nanocrystals (in the crystalline phases of *Miersite*, *Iodargyrite*) and zinc iodide nanocrystals were formed in all the synthesized HPs under the selected synthesis conditions. Also all samples of the HPs did not contain zinc oxide. This fact was established according to the X-Ray results. It was evidenced by the characteristic peaks in the X-ray diffraction patterns (fig. 4). The type of silica dioxide matrix had no impact on the likelihood of silver iodide and zinc iodide nanocrystal formation. However, a notable difference was observed. The size of the crystallites increased in the sequence SBA16, SBA15, MCM41, and MCM48.

Table 2. The average crystallite size in the AgI-ZnI₂-SiO₂ HPs.

HP sample	AgI-crystalline size, nm	ZnI ₂ -crystalline size, nm
AgI-ZnI ₂ -MCM48	24	18
AgI-ZnI ₂ -MCM41	22	16
AgI-ZnI ₂ -SBA15	21	14
AgI-ZnI ₂ -SBA16	19	13

The correlation between the linear dimensions and the type of mesoporous silica dioxide could be explained by certain principles.

First and foremost, the influence of the pores' size within the silica matrix (and consequently the interpore space) became evident. It is well-known that the MCM family has smaller pores than the SBA family. Secondly, a phenomenon referred to as "chemical pressure" manifested itself. This "chemical pressure" was essentially the influence of pH substances. The pH value of the sol was set during its synthesis by these substances. Specifically, NaOH was used in the synthesis of MCM48 particles, while NH₄OH was employed for MCM41. On the other hand, HCl was utilized for SBA16 and SBA15 particles.

The influence of the listed substances on the reaction pathway and intensity between silver nitrate, zinc nitrate, and potassium iodide varied significantly. These substances altered the pH of the solution and affected the kinetic parameters of the formation of silver iodide and zinc iodide. This effect was due to the ability of silver and zinc cations (zinc compounds are ampholytic substances in addition) to form either sparingly soluble compounds (such as silver oxide, zinc hydroxide, silver chloride, etc.) or complex compounds. Consequently, potassium iodide indirectly interacted with silver nitrate and zinc nitrate. This hypothesis will be further investigated. The formation of copper oxide could provide evidence supporting this claim.

The X-Ray results showed in figure 4. It was established that AgI and ZnI₂ nanocrystals were created in HPs, because X-ray image for those samples had specific peaks. The formation of all types of nanocrystals was for all AgI-ZnI₂-SiO₂ hybrids.

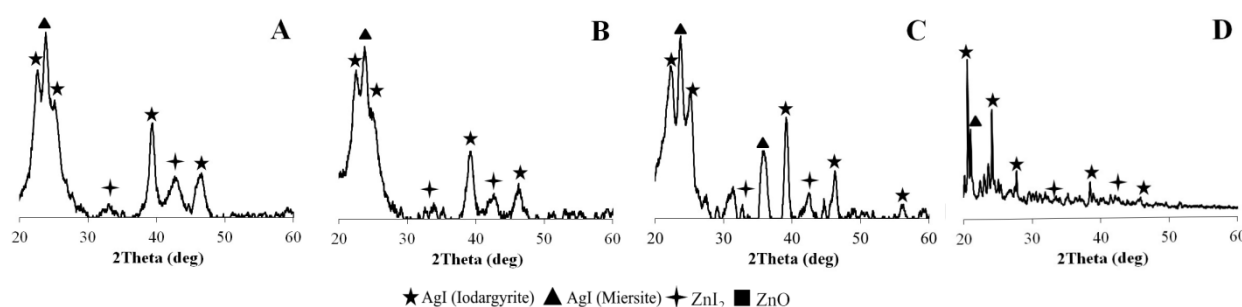


Figure 4. X-ray image of AgI-ZnI₂-SiO₂ HPs: A - AgI-ZnI₂-MCM48, B - AgI-ZnI₂-MCM41, C - AgI-ZnI₂-SBA15, D - AgI-ZnI₂-SBA16.

The AgI thermal stability was an important feature of AgI-ZnI₂-SiO₂ reagent. The AgI thermal stability was more significant than ZnI₂ thermal stability, because ZnI₂ must be destroyed under the influence of high temperatures. The thermal destruction of zinc iodide helped to prevent the silver iodide sublimation, since iodine donors were formed in the local area of combustion. The AgI thermal stability were studied by X-Ray (fig. 5), thermal gravimetric analysis (fig. 6) and atomic absorption spectroscopy with titration (table 3). The AgI thermal stability was AgI crystals preservation at the high temperature exposure at 1000 °C for 10 minutes in author's experiments. Of course, the selected mode did not fully reflect the combustion temperature of the delivery vehicle. Thus, the authors believe that the conditions for studying thermal stability were adequate, and the experimental results were reliable.

The hybrid samples underwent high-temperature treatment. X-ray analysis of these powders revealed that both silver iodide and zinc iodide remained intact in their composition. Zinc oxide was not detected in all the powders. The presence of silver iodide and zinc iodide crystals was consistently observed in all hybrid samples, irrespective of the type of silica dioxide matrix (fig. 5). This was confirmed by the characteristic peaks seen in the X-ray diffraction patterns.

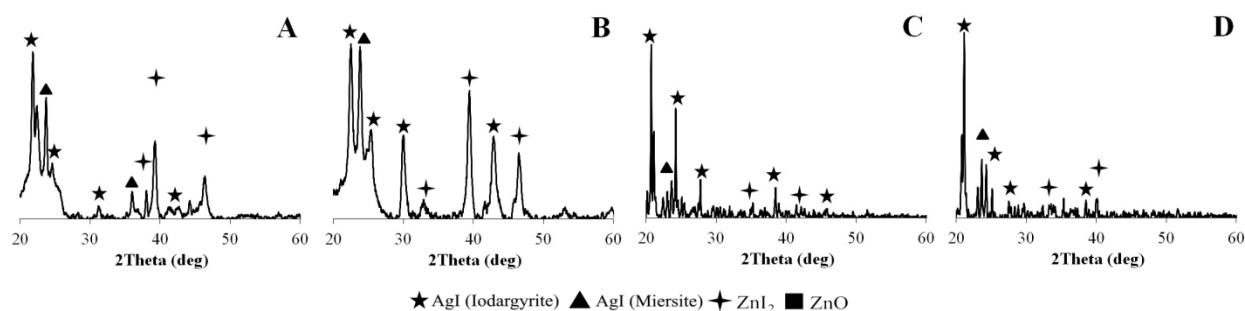


Figure 5. X-ray image of AgI-ZnI₂-SiO₂ HPs after high-temperature exposure: A - AgI-ZnI₂-MCM48, B - AgI-ZnI₂-MCM41, C - AgI-ZnI₂-SBA15, D - AgI-ZnI₂-SBA16.

Thermal stability was assessed using thermal gravimetric analysis, which revealed the dynamics of silver iodide and zinc iodide decomposition under high-temperature conditions. No qualitative or quantitative changes in the silica dioxide matrix were detected during heating. It was particularly challenging to disentangle the thermal decomposition processes of silver iodide and zinc iodide. The authors conducted experiments where AgI-SiO₂ and ZnI₂-SiO₂ hybrids, utilizing MCM48, MCM41, SBA15, and SBA16 as silica dioxide matrices, were separately exposed to high temperatures. However, these experiments did not provide a clear single influence. It was observed and confirmed that an expansion of the delayed sublimation zone (Fig. 6) occurred at temperatures ranging from 750 to 900 °C for AgI-ZnI₂-SiO₂ hybrids, compared to pure silver iodide (Fig. 6, curve E).

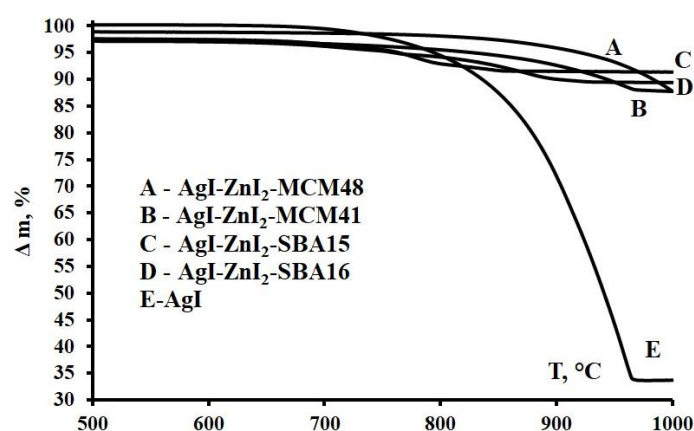


Figure 6. The thermal gravimetric curves of AgI-ZnI₂-SiO₂ HPs after high-temperature exposure.

Table 3 presents the findings from atomic absorption spectroscopy combined with titration. It was found that the type of silica dioxide matrix changed the thermal stability of silver iodide and zinc iodide. The data obtained through this method confirmed the protective screening property of the silica dioxide matrix. It was also determined that the quantity of preserved silver iodide and zinc iodide increased in the series SBA15, SBA16, MCM41 and MCM48. This phenomenon can be attributed to the silica dioxide matrix providing protection to the silver iodide and zinc iodide nanocrystals. MCM48 exhibited stronger protective properties compared to SBA15 silica dioxide due to its more complex surface structure, characterized by high structural and textural indicators.

Table 3. The average AgI content in the AgI-ZnI₂-SiO₂ HPs after high-temperature exposure.

HP sample	AgI content, %	ZnI ₂ content, %
AgI-ZnI ₂ -MCM48	71	9

AgI-ZnI ₂ -MCM41	68	7
AgI-ZnI ₂ -SBA15	66	6
AgI-ZnI ₂ -SBA16	64	6

Some of the existing exploitation characteristics were moisture absorption under static conditions and the dynamics of atmospheric moisture condensation. The results of the study were presented in tables 4.

Table 4. The moisture absorption under the static and dynamics conditions for the AgI-ZnI₂-SiO₂ HPs.

HP sample	Moisture absorption, wt. %	The duration of destruction, s
AgI-ZnI ₂ -MCM48	32	92
AgI-ZnI ₂ -MCM41	29	105
AgI-ZnI ₂ -SBA15	26	82
AgI-ZnI ₂ -SBA16	27	76

The analysis of potential operational characteristics (Tables 4) revealed that the synthesized hybrids exhibited at least two distinct mechanisms affecting atmospheric moisture. The first mechanism involved condensation due to the silica dioxide matrix, as evidenced by moisture absorption under static conditions. The second mechanism was the generation of descending air flows, as indicated by precipitation dynamics. It was observed that the condensation activity in relation to atmospheric moisture increased in the sequence MCM41, MCM48, SBA16, and SBA15. This trend was linked to the porosity of the silica matrix, with pore size playing a more significant role than surface area. Consequently, the condensation efficiency rose in proportion to the increase in mesoporous silica dioxide pore space. The intensity of the generation descending air flows was comparable across all types of AgI-ZnI₂-SiO₂ HPs. The duration of destruction for AgI-ZnI₂-MCM41 was highest, because MCM41 microagglomerates might create stable suspension in atmospheric air.

4. Conclusions

The hydrothermal template co-condensation method was employed to synthesize hybrid materials composed of AgI and ZnI₂ nanocrystals embedded within mesoporous silica dioxide, specifically utilizing MCM48, MCM41, SBA15 and SBA16 as the matrix. The resulting HPs exhibited a well-defined mesoporous structure of silica dioxide, with silver iodide and zinc iodide nanocrystals localized on the silica surface through weak physical interactions. This arrangement was achieved via the synergistic effects of the hydrothermal process and the templating capabilities of the silica dioxide matrix, facilitating the controlled deposition and stabilization of the metal halide nanocrystals.

It was established that the characterization of the silica dioxide matrix has been shown to significantly influence the morphological attributes of AgI-ZnI₂-SiO₂ HPs.

It was demonstrated that the deposition of silver iodide and zinc iodide nanocrystals onto the surface of silica dioxide matrix leads to a systematic reduction in the specific surface area of the mesoporous SiO₂. This reduction was attributable to the filling of the interpore space and pore structures with the deposited nanocrystals, effectively occluding the available surface area. The extent of this reduction in specific surface area varied across different SiO₂ matrix as evidenced by the following data: 40 m²/g for MCM48, to 33 m²/g for MCM41, to 27 m²/g for SBA15, to 25 m²/g for SBA16. Furthermore, the analysis revealed significant alterations in the pore volume metrics and linear dimensions of the SiO₂ matrix, irrespective of the matrix type. These findings underscore the profound impact of nanocrystal deposition on the porous structure of silica dioxide, highlighting the need for a comprehensive understanding of the physicochemical processes involved in such transformations. The observed changes in pore characteristics suggested that the crystallization

process did not only modifying the surface area but also affected the internal architecture of the SiO₂ matrix.

It has been shown that the characterization of the structure of mesoporous silicon dioxide did not affect the growth and morphology of silver iodide and zinc iodide nanocrystals. However, the properties of the matrices and the pH-forming substances, which were used for synthesis, had an indirect effect on crystal formation.

It has been found that the morphology and physico-chemical properties of mesoporous silicon dioxide play a key role in the formation of the final morphology and sizes of metal halide nanocrystals.

The thermal stability of silver iodide and zinc iodide was demonstrated to significantly enhance due to the protective shielding effect of the silica dioxide matrix. The analysis revealed that after high-temperature exposure, the silver iodide content varied across different mesoporous silica dioxide matrix: specifically, 71 % for MCM48, 68 % for MCM41, 66 % for SBA15, and 64 % for SBA16. This observation underscored the critical role of the SiO₂ matrix in preserving the integrity and stability of the iodide-based species under high temperature.

The analysis of the AgI-ZnI₂-SiO₂ operational characteristics revealed differences, which provide a basis for the theoretical classification of novel reagents designed for artificial manipulation of atmospheric moisture. These differences could be potentially utilized to optimize their application under diverse atmospheric conditions, including varying cloud water content, altitude, and other relevant meteorological factors. This approach did not only enhance the precision of atmospheric engineering but also broadens the applicability of these reagents in a wide range of environmental contexts.

The synthesized reagent demonstrated significant potential to enhance the efficacy of artificial interventions in meteorological phenomena. Its implementation is expected to optimize the modulation of atmospheric dynamics, thereby improving the controllability of dangerous weather phenomena.

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Competing interests: The authors declare that they have no competing interests.

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